

A One-Pass, Collapse-Resistant Neural Topic Model for Mass2Motif Discovery

Abstract

Mass2Motif discovery requires a model to explain held-out spectral evidence while retaining a broad inventory of chemically interpretable fragment and neutral-loss patterns. We present a neural probabilistic topic model in which one sparse top-2 routing pass replaces iterative document-level inference. Fixed, train-only token features and learned topic prototypes define both the router and the topic-word decoder. Mean-normalized fragment/loss evidence removes a vocabulary-size bias, while balanced assignments, paired views, positive-NPMI structure, prototype separation, and dead-topic recycling address topic collapse. We compare the model with Tomotopy LDA using the same 1000 topics, seed 42, 6 CPU threads, scaffold split, and leakage-filtered MAG protocol. On validation, the neural model produces 663 optimized motifs and 185 useful high-confidence motifs, versus 607 and 138 for Tomotopy. On test, the corresponding useful-motif counts are 234 and 186. Tomotopy retains higher mean structure overlap score (SOS), so the result is a breadth-purity trade-off rather than uniform superiority. Fitting times are similar (78.7 versus 76.0 minutes). This is a single-dataset, single-seed research comparison, not a claim that the neural model should replace the production backend.

1 Introduction

MS2LDA represents tandem mass spectra as documents, fragment and neutral-loss tokens as words, and recurrent co-occurrence patterns as Mass2Motifs [1]. A useful large-scale model must do more than minimize a predictive loss: it should retain many distinct motifs, assign spectra confidently enough for motif-level analysis, and support chemical annotation [2].

Conventional latent Dirichlet allocation (LDA) estimates a new document’s topic mixture by iterative local inference [7]. Amortized neural inference can replace that loop with a learned mapping, but a model with 1,000 topics can collapse: a small subset of topics absorbs most probability mass, several prototypes become near-duplicates, or learned topics ignore observed token co-occurrence. Mass-spectrometry vocabularies introduce another bias. If fragment and neutral-loss channel evidence is summed, the larger channel gains mass simply because it contains more tokens, even when its mean logit is no better.

We address these problems with one shared prototype geometry. It defines a fragment/loss-balanced topic-word distribution and a contextual top-2 router whose aggregated assignments form the document mixture in one pass. The training objective combines cross-view prediction with complementary safeguards against usage, geometric, and co-occurrence collapse. This report states the complete model and evaluation protocol directly; a reader should not need the preceding architecture experiments to reproduce the method.

2 Methods

2.1 Notation and data representation

Let $d \in \{1, \dots, D\}$ index spectra, $w \in \{1, \dots, V\}$ index the shared fragment/loss vocabulary, and $k \in \{1, \dots, K\}$ index topics. The count $c_{dw} \geq 0$ is the historical MS2LDA token multiplicity. The model returns a document-topic distribution $\theta_d \in \Delta^{K-1}$ and a topic-word distribution $\beta_k \in \Delta^{V-1}$, with

$$p(w \mid d) = \sum_{k=1}^K \theta_{dk} \beta_{kw}. \quad (1)$$

The public inputs are distributed in Zenodo record 20179680 [9]. The source MGF contains 41568 spectra; deterministic cleaning retains 38888. Molecules are keyed by the connectivity block of the InChIKey. Cyclic molecules are grouped by non-chiral Bemis-Murcko scaffold [3], while each acyclic connectivity group remains intact. Groups are assigned by descending size to 70/10/20 train, validation, and test targets using seed 42. The resulting matrices contain 27222 training, 3889 validation, and 7777 test spectra, with no compound or scaffold overlap.

Peak intensities are divided by the spectrum maximum. Peaks outside $m/z \in (0, 2000]$ or normalized intensity $[0.01, 1]$ are discarded; at most 1,000 peaks are retained, and a spectrum must retain at least five. Each peak yields a fragment token and, when positive, a precursor-minus-peak neutral-loss token, both rounded to two decimal places. A token at normalized intensity I occurs $\text{round}(100I)$ times. The vocabulary is constructed only from training spectra, requires document frequency at least three, preserves first-seen order, and contains 21233 tokens.

For held-out completion, whole physical peak groups are divided deterministically 50/50. A fragment and its corresponding loss therefore cannot occur on opposite sides. The observed half is renormalized without consulting the withheld maximum. Model development uses only training and validation. Test evaluation is called once after validation evaluation and MAG scoring are complete.

2.2 Fixed token features

Each token has a fixed feature vector $x_w \in \mathbb{R}^{64}$. Its first 48 coordinates are skip-gram negative-sampling (SGNS) embeddings trained only on training documents [4]. For five epochs, eight count-weighted positive pairs are sampled per document; each positive pair uses five negatives drawn with probability proportional to corpus frequency raised to 0.75. Source embeddings are initialized uniformly on $[-0.5/48, 0.5/48]$ and context embeddings at zero. SparseAdam uses learning rate 0.01 and batches of 8,192 pairs. Source and context embeddings are averaged and normalized after the fifth epoch.

If m_w denotes token mass, fourteen Fourier coordinates encode mass at $f \in \{1, 2, 4, 8, 16, 32, 64\}$:

$$\phi_f(m_w) = [\sin(2\pi f m_w / 1000), \cos(2\pi f m_w / 1000)]. \quad (2)$$

The Fourier block is divided by $\sqrt{7}$. Two final one-hot coordinates mark fragment versus neutral loss. Concatenating the normalized SGNS vector, Fourier block, and type indicators gives 64 coordinates; the complete vector is normalized again. Molecular labels and pretrained spectrum encoders are not used.

2.3 Shared topic geometry and balanced decoder

A learned bias-free matrix $W \in \mathbb{R}^{128 \times 64}$ projects the fixed features, and every topic has a learned prototype $q_k \in \mathbb{R}^{128}$:

$$z_w = \text{normalize}(W x_w), \quad \hat{q}_k = \text{normalize}(q_k). \quad (3)$$

The matrix W is initialized orthogonally. The same z_w and $\hat{q}_k$ are used by both decoder and router. Decoder logits are temperature-scaled cosine similarities,

$$\ell_{kw} = \frac{2\hat{q}_k^\top z_w}{\tau_\beta}, \quad \tau_\beta = 0.18. \quad (4)$$

Let V_F and V_L denote the fragment and neutral-loss vocabularies. Channel evidence is a log mean exponential, not a log sum exponential:

$$e_{kt} = \text{logsumexp}_{w \in V_t}(\ell_{kw}) - \log |V_t|, \quad \pi_{kt} = \text{softmax}_{t \in \{F, L\}}(e_{kt}). \quad (5)$$

The subtraction removes the $\log |V_t|$ multiplicity advantage: if all token logits are equal, the two channels receive equal evidence even when their vocabularies differ in size. A fixed $\lambda = 0.25$ pulls the inferred channel mass toward one half,

$$\tilde{\pi}_{kt} = (1 - \lambda)\pi_{kt} + \frac{\lambda}{2}, \quad (6)$$

and the row-stochastic decoder is

$$\beta_{kw} = \tilde{\pi}_{k,t(w)} \frac{\exp(\ell_{kw})}{\sum_{v \in V_{t(w)}} \exp(\ell_{kv})}. \quad (7)$$

Thus $\sum_w \beta_{kw} = 1$, rankings within each channel are unchanged, and the channel correction adds no learned parameter.

2.4 Contextual top-2 router and document mixture

For spectrum d , define the count-weighted embedding sum $s_d = \sum_w c_{dw} z_w$. For each non-zero token entry, the leave-one-out context is

$$\bar{z}_{d \setminus w} = \frac{s_d - c_{dw} z_w}{\max(\sum_v c_{dv} - c_{dw}, 1)}. \quad (8)$$

A two-layer MLP (linear, GELU, layer normalization, linear) with a 256-unit hidden layer gives

$$h_{dw} = \text{normalize}(z_w + \text{MLP}([z_w; \bar{z}_{d \setminus w}])), \quad u_d = \text{normalize}(s_d). \quad (9)$$

Local and document evidence are added before routing:

$$a_{dwk} = h_{dw}^\top \hat{q}_k + u_d^\top \hat{q}_k, \quad r_{dw}^{\text{soft}} = \text{softmax}_k(a_{dwk}/T). \quad (10)$$

Only the two largest entries of r_{dw}^{soft} are retained and renormalized to form r_{dw}^{hard} . Training uses the straight-through quantity $r^{\text{hard}} + r^{\text{soft}} - \text{stopgrad}(r^{\text{soft}})$; inference uses r^{hard} directly.

Count-weighted routed mass is $M_{dk} = \sum_w c_{dw} r_{dwk}$. The same whole-spectrum score supplies a multiplicative document gate,

$$g_{dk} = \text{softmax}_k(u_d^\top \hat{q}_k/T), \quad \theta_{dk} = \frac{M_{dk} g_{dk}^\gamma}{\sum_j M_{dj} g_{dj}^\gamma}, \quad \gamma = 0.75. \quad (11)$$

The gate is detached in Equation 11 during training, so it cannot reduce the loss by sharpening itself; its document score still learns through Equation 10. Multiplication preserves exact zero support from token routing. If a spectrum has no routed mass, θ_d is explicitly uniform. Equations 10–11 require one forward pass and no per-spectrum optimization loop.

The count-weighted completion loss implements Equation 1 exactly at non-zero target entries:

$$\mathcal{L}_{\text{comp}}(\theta, X) = -\frac{1}{\sum_{dw} c_{dw}} \sum_{dw} c_{dw} \log \left(\sum_k \theta_{dk} \beta_{kw} \right). \quad (12)$$

Sparse indexing avoids materializing the dense matrix $\theta\beta$ but does not change its probabilities.

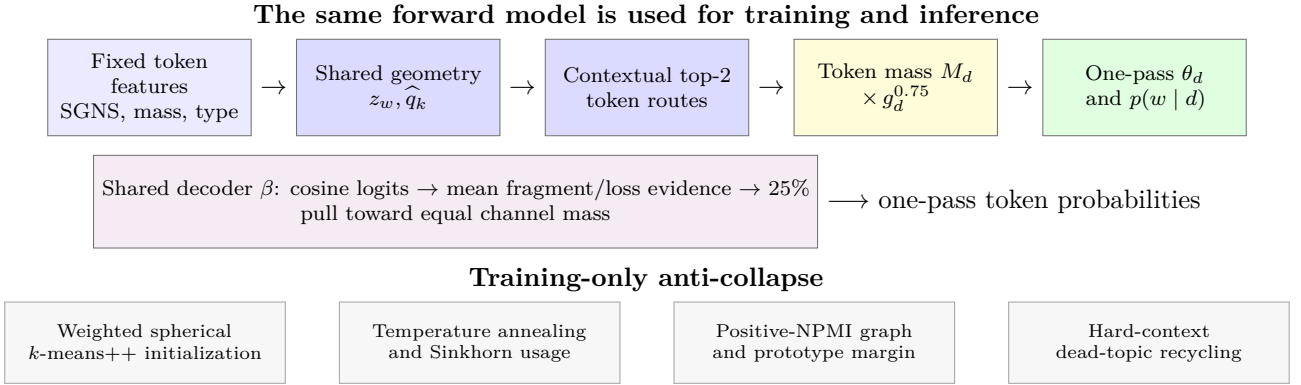


Figure 1: The implemented model. Fixed train-only features and learned prototypes define both the one-pass router and balanced decoder. The lower row contains training-only safeguards and is absent from inference.

2.5 Alternating paired-view objective

Four deterministic view pairs are constructed from each training spectrum. Each side retains 80 percent of whole physical peak groups (while withholding at least one when possible) and is renormalized independently. Epochs cycle through the four pairs.

Optimization alternates two gradient blocks using one AdamW optimizer. In the *router block*, β is cached and detached. Each partial view predicts the other, while Sinkhorn targets and Jensen–Shannon agreement regularize the router:

$$\begin{aligned} \mathcal{L}_{\text{router}} = & \frac{1}{2} [\mathcal{L}_{\text{comp}}(\theta^A, X^B) + \mathcal{L}_{\text{comp}}(\theta^B, X^A)] + \omega_S \mathcal{L}_{\text{Sinkhorn}} \\ & + 0.1 \text{JS}(\theta^A, \theta^B) + 5\mathcal{L}_{\text{sep}}. \end{aligned} \quad (13)$$

For a mini-batch with N routed entries, log-space Sinkhorn iteration projects $\exp(a/\varepsilon)$, with $\varepsilon = 0.05$, to detached targets Q whose rows sum to one and whose columns sum to N/K [5, 6]. $\mathcal{L}_{\text{Sinkhorn}}$ is cross-entropy from Q to $\text{softmax}(a/T)$. Sinkhorn is a training target only; it is not part of held-out inference.

In the *topic block*, routes are recomputed without gradients and held fixed. Decoder geometry is then optimized with

$$\mathcal{L}_{\text{topic}} = \mathcal{L}_{\text{comp}} + 0.25\mathcal{L}_{\text{local}} + \mathcal{L}_{\text{NPMI}} + 5\mathcal{L}_{\text{sep}}, \quad (14)$$

where

$$\mathcal{L}_{\text{local}} = -\frac{1}{\sum_{dw} c_{dw}} \sum_{dw} c_{dw} \sum_k \text{stopgrad}(r_{dwk}) \log \beta_{kw}. \quad (15)$$

This term asks each routed topic to emit the token assigned to it. A full router pass with batch size 128 is followed by four topic-block updates with batch size 512 per epoch.

2.6 Anti-collapse design

The safeguards target different collapse modes; removing one is not equivalent to strengthening another.

1. **Dispersed initialization.** Topic prototypes use deterministic spherical k -means++ seeding without Lloyd updates. Token sampling weight is $\sqrt{\text{CF}_w} \text{IDF}_w^2$, where $\text{IDF}_w = \log[(1+D)/(1+\text{DF}_w)]$. This retains common evidence while favouring distinctive tokens.
2. **Annealed sparse routing.** Routing temperature falls linearly from 0.5 to 0.1 over 30 epochs. Early soft assignments permit exploration; later top-2 assignments specialize.
3. **Balanced usage.** The Sinkhorn coefficient is 0.25 for the first ten epochs and then decreases linearly to 0.05 at epoch 40. It resists early winner-take-all collapse without forcing uniform usage at the end of training.
4. **Train-only co-occurrence structure.** Binary training documents define a positive-NPMI graph. Tokens require document frequency at least ten, pair frequency at least three, and mutual membership among at most 16 strongest neighbours. If A is the resulting weighted sparse adjacency, the retained edge weight is

$$A_{wv} = \frac{\log[p(w, v)/(p(w)p(v))]}{-\log p(w, v)}, \quad A_{wv} > 0, \quad (16)$$

where probabilities are empirical document frequencies. Only mutual top-neighbour edges remain. The training loss is

$$\mathcal{L}_{\text{NPMI}} = -\frac{1}{K} \sum_k \log \left(\sum_{wv} \beta_{kw} A_{wv} \beta_{kv} \right). \quad (17)$$

This favours probability mass on training-supported token pairs.

5. **Prototype separation.** For every prototype, the eight nearest other prototypes are penalized above cosine margin 0.3:

$$\mathcal{L}_{\text{sep}} = \text{mean}_{k,j} \left[\max(0, \hat{q}_k^\top \hat{q}_j - 0.3) \right]^2. \quad (18)$$

This detects geometric duplication that usage balance cannot detect.

6. **Dead-topic recycling.** Every second epoch, validation full-spectrum usage is measured. A topic below $0.1/K$ mean usage for three consecutive checks is replaced by a stored high-loss routing context. The bounded queue retains at most $4K$ contexts and admits the 32 highest-loss contexts from each routed view and mini-batch. The replaced prototype’s Adam state is cleared. A topic can be recycled at most twice; 28 replacements occurred in this run.
7. **Numerical controls.** Non-finite losses or gradients stop training immediately, and gradient norm is clipped to 5.0. Training always runs to the predeclared epoch 40; no validation-metric model search is performed. AdamW uses learning rate 10^{-3} , weight decay 10^{-4} , and seed 42. Table 3 lists the complete settings.

2.7 Comparator and evaluation

Tomotopy 0.13.0 [8] is trained independently on the same training matrix and never acts as a teacher. It uses $K = 1000$, scalar initial $\alpha = 0.6$ with optimization interval 10, $\eta = 0.1$, seed 42, 6 workers, and parallel scheme 3. Training advances in blocks of 50 to at most 5,000 iterations and stops when all of the last ten relative perplexity changes are below 0.005. The recorded fit stops at 750 iterations. Held-out Tomotopy mixtures use 100 iterations, 6 workers, parallel scheme 1, and independent-document inference. The neural model has no corresponding iteration count because its document mapping is amortized into one routing pass.

Reported fitting time is optimization-loop wall time. For the neural model it excludes MGF processing, SGNS feature construction, co-occurrence graph construction, initialization, and evaluation; for Tomotopy it excludes sparse matrix expansion, model construction, and evaluation. This gives like-for-like core fitting time but is not total workflow time.

Predictive evaluation is count-weighted completion NLL per in-vocabulary token. Chemical evaluation follows the paper-aligned MAG workflow [2]. Before retrieval, all reference entries sharing a validation or test connectivity key are removed. The 20 most probable tokens of each topic form a motif spectrum. MAG retrieves up to 1,000 neighbours, retains ten unique molecules, clusters at cosine similarity 0.9, and optimizes motif features. A topic with at least one optimized feature counts toward annotation coverage.

A high-confidence topic-spectrum association uses the full-spectrum mixture and requires $\theta_{dk} \geq 0.5$. For an annotated topic, MACCS bits present in at least 80% of its clustered MAG molecules define binary fingerprint A_k . For associated compound i with fingerprint B_i , the containment score is

$$s_{ki} = \frac{|A_k \cap B_i|}{|A_k|}, \quad \text{SOS}_k = \frac{1}{|C_k|} \sum_{i \in C_k} s_{ki}, \quad (19)$$

where C_k contains unique connectivity keys, preventing repeated spectra of one molecule from dominating. A topic is SOS-evaluable only when MAG retained at least one optimized feature and C_k contains an associated

molecule with a valid fingerprint. Among SOS-evaluable high-confidence motifs, bands are high (> 0.8), intermediate ($0.6 \leq \text{SOS} \leq 0.8$), and low (< 0.6). A *useful high-confidence motif* lies in the high or intermediate band. Optimized, evaluable, and useful motif counts, SOS bands, mean/median SOS, and fitting time form the primary comparison. NLL, collapse checks, and warm resident inference are secondary diagnostics.

3 Results

3.1 Validation comparison

Table 1: Primary validation comparison.

Metric	Neural	Tomotopy
Optimized motifs	663	607
MAG annotation coverage	66.3%	60.7%
High-confidence SOS-evaluable motifs	312	206
SOS high (> 0.8)	49	58
SOS intermediate (0.6–0.8)	136	80
SOS low (< 0.6)	127	68
Useful motifs (high-confidence; $\text{SOS} \geq 0.6$)	185	138
Mean SOS (high-confidence)	0.6323	0.6761
Median SOS (high-confidence)	0.6364	0.6854
Model-fitting time (minutes)	78.7	76.0

The neural model yields 663 optimized motifs versus 607 for Tomotopy, and 312 high-confidence SOS-evaluable motifs versus 206. Useful-motif counts are 185 and 138, respectively. The broader neural inventory has lower mean SOS (0.632 versus 0.676). Model-fitting times are close: 78.7 minutes for neural MS2LDA and 76.0 minutes for Tomotopy.

3.2 Test confirmation

Table 2: Primary test comparison after validation-only development.

Metric	Neural	Tomotopy
Optimized motifs	663	607
MAG annotation coverage	66.3%	60.7%
High-confidence SOS-evaluable motifs	424	333
SOS high (> 0.8)	58	66
SOS intermediate (0.6–0.8)	176	120
SOS low (< 0.6)	190	147
Useful motifs (high-confidence; $\text{SOS} \geq 0.6$)	234	186
Mean SOS (high-confidence)	0.6204	0.6369
Median SOS (high-confidence)	0.6135	0.6145
Model-fitting time (minutes)	78.7	76.0

The test result preserves the validation pattern. Neural MS2LDA yields 234 useful high-confidence motifs versus 186, and 424 SOS-evaluable motifs versus 333. Annotation coverage is 66.3% versus 60.7%. Mean SOS remains lower for the neural model (0.620 versus 0.637); the corresponding medians are reported in Table 2.

4 Discussion

Mean-normalized type evidence resolves a specific probabilistic bias rather than adding network capacity. Log-sum-exp channel evidence contains a $\log |V_t|$ term; removing it makes fragment and neutral-loss evidence comparable before the fixed equal-mass pull. Because the correction introduces no learned parameter, it is less dataset-specific than increasing encoder size or adding another loss.

Parity is metric-dependent. The neural model produces more optimized, high-confidence evaluable, and useful motifs, predicts withheld tokens with lower NLL (Appendix C), and fits in similar time. Tomotopy has higher mean validation and test SOS. The neural result is therefore broader, not uniformly purer. A single seed

was used deliberately to improve and inspect one architecture without averaging away its failure modes, but this design does not measure uncertainty across initializations or datasets.

The most consequential future extension is a jointly trained spectrum-level encoder. The current u_d is a normalized sum of projected token embeddings; a set or transformer encoder could represent global peak relations that a sum cannot, while the prototype decoder retains interpretable motifs. Other natural extensions are sparse chemical supervision where reliable labels exist, fine-tuning on a new dataset, GPU-native sparse batches, and optionally freezing known motif prototypes while other topics learn. Selective freezing is useful but is not unique to neural inference because variational-Bayes LDA can also fix chosen motifs. None of these extensions contributes to the reported result.

The limitations are concrete. This is one dataset and one seed. MAG outcomes depend on the retrieval library, clustering, and motif optimization. Warm inference excludes parsing, feature construction, model loading, and cold-start costs. Finally, topic counts and SOS characterize the evaluable motif inventory; they do not prove that every topic corresponds to a distinct biochemical substructure.

5 Conclusion

A shared prototype geometry, one-pass contextual routing, mean-normalized fragment/loss evidence, and complementary anti-collapse training produce a broad neural Mass2Motif inventory with strong predictive fit. Under the fixed seed-42, $K = 1000$, 6-thread protocol, neural MS2LDA exceeds Tomotopy in optimized and useful motif counts, while Tomotopy retains higher mean SOS. The next scientific step is replication across datasets and seeds, not further adaptation to the evaluated test split.

A Exact hyperparameters

Table 3: Values generated from `protocol.json`.

Quantity	Value
Seed; topics; CPU threads	42; 1000; 6
Token features	48 SGNS + 14 Fourier + 2 type = 64 dimensions
SGNS	SparseAdam; 5 epochs; 8 pairs/document; 5 negatives; power 0.75; batch 8192; lr 0.01
Projection; router hidden	128; 256
Decoder	temperature 0.18; mean channel evidence; equal-mass pull 0.25
Router	top-2; additive document evidence; gate exponent 0.75
Views	4 pairs; 80% physical peak groups/view
AdamW	lr 0.001; weight decay 0.0001; gradient clip 5.0
Batches and epochs	router 128; topic 512; 4 topic updates/epoch; 40 epochs
Loss weights	local 0.25; view consistency 0.1; NPMI 1.0; separation 5.0
Routing temperature	0.5 to 0.1 over 30 epochs
Sinkhorn	epsilon 0.05; 5 iterations; weight 0.25 to 0.05
NPMI graph	min DF 10; min pair 3; 16 neighbours; positive NPMI
Prototype separation	8 neighbours; margin 0.3
Recycling	usage < 0.1 of uniform for 3 validations; max 2/topic
Tomotopy	alpha 0.6; eta 0.1; parallel 3; step 50; max 5000; stop 10 changes < 0.005; inference 100
MAG/SOS	top 20 motif tokens; search 1000; 10 unique molecules; membership ≥ 0.5 ; cluster cosine 0.9; fingerprint consensus 0.8

B Equation–implementation correspondence

Table 4: Direct map from the mathematical description to executable code.

Report component	Implementation
Token features x_w	<code>data.py: build_token_features</code>
Projected tokens z_w	<code>model.py: projected_tokens</code>
Decoder β	<code>model.py: topic_word_distribution</code>
Leave-one-out context	<code>model.py: _route_embeddings</code>
Top-2 routing	<code>model.py: route</code>
Gated θ_d	<code>model.py: aggregate_theta</code>
Likelihood and regularizers	<code>objectives.py</code>
Optimization and recycling	<code>training.py; optimization.py</code>
Held-out inference	<code>evaluation.py</code>
MAG and SOS	<code>mag.py; chemical.py</code>

C Secondary diagnostics and practical inference

On test, 319 topics exceed uniform mean corpus usage and the median entropy-derived effective topic count is 7.28 per spectrum. These quantities are collapse checks, not headline chemical outcomes. Completion NLL per token is 8.5014 versus 9.6622 on validation and 8.5226 versus 9.7569 on test (neural versus Tomotopy).

In warm in-memory inference on resident models and batches of 128 spectra, the neural router processes 6554.4 spectra/s versus 30.0 spectra/s for Tomotopy, a 218.1-fold ratio. Both measurements use 6 CPU threads; the neural model uses one routing pass and Tomotopy uses 100 inference iterations. This result is relevant to repeated in-memory batches or a resident service, not end-to-end application latency.

D Executable reproduction

The repository contains one current protocol and one current model artifact. The latter consists only of the model state, vocabulary, and the two inference temperatures. From the repository root:

```
conda env create -f environment-neural-ms2lda.yml
conda env create -f environment-msnlib-mag.yml
conda run -n ms2lda-neural python \
  scripts/download_msnlib_validation_assets.py \
  --data-root /path/to/MSnLib-assets
conda run -n ms2lda-neural python -m benchmarks.neural_ms2lda run \
  --data-root /path/to/MSnLib-assets --run /path/to/run
python scripts/generate_neural_ms2lda_report.py
```

The acquisition command validates the public source data before training. Validation evaluation and MAG scoring finish before test evaluation is invoked. The comparator is trained from the same training matrix, so no external Tomotopy result is required.

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